

OE-EC804A — Artificial Intelligence

Semester VIII • 3L:0T:0P • 3 Credits

Module 6: First-Order Logic (FOL)

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Syntax and Semantics of First-Order Logic

First-Order Logic (FOL) extends propositional logic by representing objects, relations, and functions explicitly.

Elements of First-Order Logic

- **Objects:** Concrete entities in the world (e.g., *John*, *5*, *CGEC*, *Wumpus*).
- **Relations / Predicates:** Properties or connections between objects (e.g., *Brother(John, Richard)*, *Greater(5, 3)*).
- **Functions:** Operations mapping objects to another object (e.g., *MotherOf(John) = Mary*).
- **Quantifiers:** $\forall$ (Universal: "for all") and $\exists$ (Existential: "there exists").

1.1 Quantifiers in First-Order Logic

- **Universal Quantifier ($\forall$):** $\forall xP(x)$ means $P(x)$ is true for all objects x in the domain. *Example:* "All students are smart" $\rightarrow \forall x(Student(x) \Rightarrow Smart(x))$.
- **Existential Quantifier ($\exists$):** $\exists xP(x)$ means $P(x)$ is true for at least one object x in the domain. *Example:* "Some student is smart" $\rightarrow \exists x(Student(x) \wedge Smart(x))$.

Quantifier Duality Rules

$$\forall x \neg P(x) \equiv \neg \exists x P(x) \quad ; \quad \neg \forall x P(x) \equiv \exists x \neg P(x) \quad (1)$$

2. Unification & First-Order Inference

Definition of Unification

Unification is an algorithmic process of finding a substitution θ that makes two first-order logic expressions identical.

Unification Function

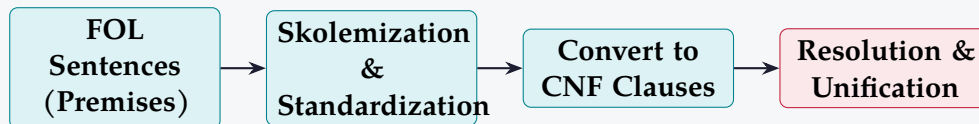
$$Unify(p, q) = \theta \quad \text{such that } SUBST(\theta, p) = SUBST(\theta, q) \quad (2)$$

where θ is the **Most General Unifier (MGU)**.

2.1 Example of Unification

Suppose we want to unify $Knows(John, x)$ and $Knows(John, Jane)$:

$$Unify(Knows(John, x), Knows(John, Jane)) = \{x/Jane\} \quad (3)$$

First-Order Logic Resolution Pipeline

Quick Revision — Module 6

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Key Concepts and Metrics at a Glance

Universal Quantifier	$\forall x(P(x) \implies Q(x))$ (True for all objects in domain)
Existential Quantifier	$\exists x(P(x) \wedge Q(x))$ (True for at least one object in domain)
Quantifier Duality	$\neg \forall x P(x) \equiv \exists x \neg P(x)$; $\neg \exists x P(x) \equiv \forall x \neg P(x)$
Unification Function	$SUBST(\theta, p) = SUBST(\theta, q)$ (θ is Most General Unifier)
Skolemization	Replaces existential variable $\exists x$ with Skolem constant/function

One-Line Definitions

- **Predicate:** A logical function returning True/False based on object relations.
- **Unification:** Algorithm that computes variable substitutions to make two expressions match.
- **Skolemization:** Process of replacing existential variables with Skolem constants/functions.
- **MGU (Most General Unifier):** The simplest substitution that unifies two logic expressions.

Frequently Asked / Viva Questions

Q1. Compare Propositional Logic and First-Order Logic.

Propositional logic assumes world contains atomic facts. FOL assumes world contains objects, relations, functions, and quantifiers.

Q2. Explain the use of Universal Quantifier ($\forall$).

Expresses that a property holds for all objects in the domain. Typically used with implication ($\implies$).

Q3. Explain the use of Existential Quantifier ($\exists$).

Expresses that a property holds for at least one object in domain. Typically used with conjunction ($\wedge$).

Q4. What is Unification in FOL?

Finding a substitution set θ that makes two predicate expressions identical.

Q5. What is Skolemization?

Eliminating existential quantifiers $\exists x$ by substituting x with a Skolem constant or function depending on outer universal variables.

Frequently Asked / Viva Questions (contd.)**Q6. State the Quantifier Duality rule.**

$\neg\forall xP(x) \equiv \exists x\neg P(x)$ and $\neg\exists xP(x) \equiv \forall x\neg P(x)$.

Q7. Why is $\wedge$ usually incorrect with $\forall$?

$\forall x(Student(x) \wedge Smart(x))$ asserts that *everything in the universe* is a student and is smart! $\Rightarrow$ should be used instead.

Q8. What is Generalized Modus Ponens?

An inference rule in FOL that combines Modus Ponens with Unification to infer conclusions directly from KB.

Q9. What is a Most General Unifier (MGU)?

A unifier that places the fewest restrictions on variables.

Q10. What is Ground Resolution?

Resolution applied to ground (variable-free) sentences.